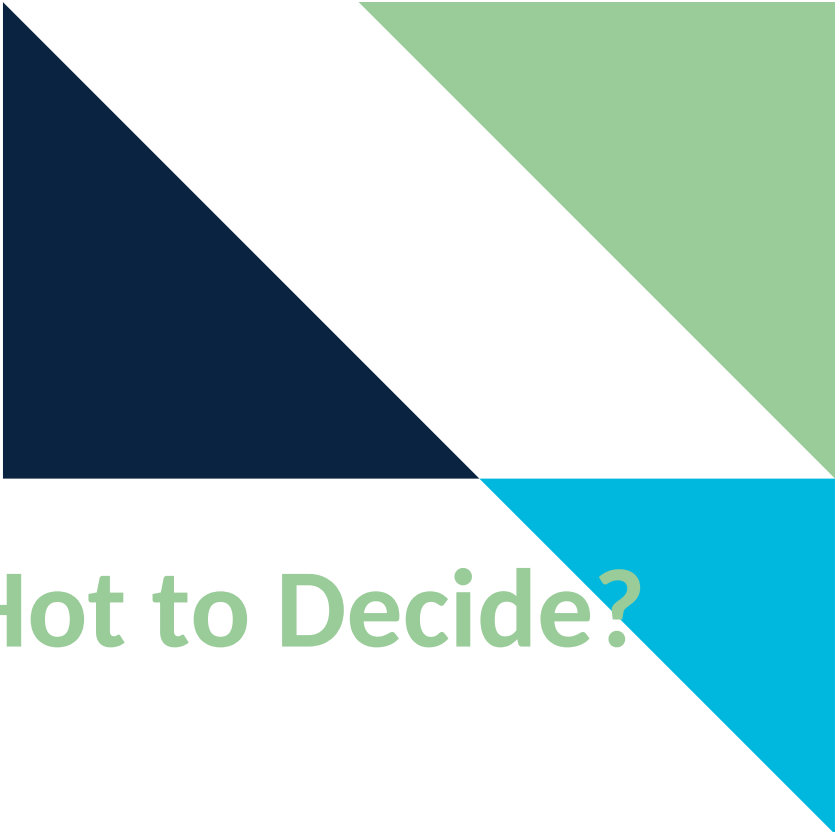


IMT Atlantique

PRONTO Project – S6 2025-2026

Nantes Campus



PRONTO: Hot to Decide?

Project no. 8, team 1

FINAL REPORT

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1. Introduction

The project “**Chaud pour décider ?**” (**Hot to Decide?**) addresses today’s challenges of energy transition and sustainable land planning. Faced with the climate emergency, small municipalities must meet a twofold challenge: increase the share of renewable energy in their energy mix while supporting ever denser urban development.

In this context, building a centralized urban heating network appears to be a relevant collective solution to guarantee more frugal, local and sustainable energy production and distribution.

Our problem statement is therefore to develop a decision-support tool designed to optimize the layout of the heating network across the town. This tool aims to help technical decision-makers define an efficient network architecture, able to reconcile the economic viability of the project, the social objectives of the municipality and the environmental requirements of the energy transition. The system must therefore propose an optimized network configuration that meets technical, economic and usage-distribution criteria.

Indeed, even though the main objective of this project is to obtain a minimum operating cost, we must also make sure we use 80-90% of the power supplied by our heat source, connect a significant share of households, and produce a layout map that non-experts can read.

The project was carried out in two parts:

- **Part one:** characterized by a single heat source in the network, whose maximum power is known in advance. The objective is to build an optimal heating network that respects the constraints stated above, by translating them into a linear mathematical problem.
- **Part two:** several source nodes with different positions and delivered power. The objective of this phase is to determine the source (size and coordinates) and the optimal network while respecting the constraints defined in the specification. This requires a metaheuristic algorithm whose parameters must be chosen carefully.

2. Feedback on our functional specification and mathematical modelling

Function	Criterion/criteria	Level(s)	Flexibility	Priority
MF1 – Build a fair heating network				
MF1.1 – Minimize costs	Network optimization	Lowest amount: if	High	1

		possible below €0.08/kWh		
MF1.2 - Cost-effectiveness of spending	Do not connect every building	Minimum cost while respecting the constraints: minimize the investment cost per kWh	Medium	2
MF2 - Visualization of the result				
MF2.1 - The result must be usable by the people in charge of building the network.	Result understandable by the people in charge of building the network	Approval by the project manager	None	1
CF1 - Respect the connection constraints				
CF1.1 - Share of connected buildings	Connect a sufficient number of buildings	50% of buildings connected	Medium	2
CF1.2 - Use of the heat source	Do not consume (losses included) more than the heat source allows during a demand peak.	The maximum production (3,500 kW) must never be exceeded by the demand	None	2
CF1.3 - Mix of connected building types	Respect the proportions of connected public services, housing and businesses.	30% housing 30% public services 20% businesses	Medium	2
CF1.4 - Validation of the need	Meet the stated need	Model validated by the town hall	None	1
CF2 - Legislation				
CF2.1 - Compliance with the law	The system must comply with the laws in force.	Validation by the supervisor	None	3
CF3 - Environment				
CF3.1 - Respect for the environment	Propose a solution that uses few computing	Follow green coding rules	Medium	4

	resources			
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Table 1. Table of functions, with their value criterion, level, flexibility and priority. MF: main function, CF: constraint function.

2.1 Part one

- A. Mathematical model: (see Appendix 2)
- B. Match with the specification:

To meet the needs expressed in the specification, we chose to rely on a mathematical model that allows us to validate it. To reach the minimum-cost objective, there is an objective function to minimize whose terms account for all the characteristics of the heating network, in particular the losses, which must be minimized. We can therefore state that we respected all the constraints set by the town hall while minimizing costs.

In addition, to obtain a readable and understandable graph, we chose to use the “*NetworkX*” library. As required by the specification, the visualization was approved by the town hall.

To respect the equity constraint, we came up with three different ways to model it: the first based on building types, the second on meeting a demand threshold, and the third on satisfaction according to floor surface. After running our code with these three models (see Results), we finally decided to keep a single approach, the one based on surfaces. In the context of a town, it seemed more relevant to us to ensure a fair distribution at the scale of “neighbourhoods” rather than according to demand alone, in order to guarantee an equity that is more consistent with and representative of the town’s structure.

Finally, regarding environmental and legal compliance, in this part we did not have access to data that would allow us to check the environmental and regulatory dimensions rigorously.

2.2 Part two

- A. Mathematical model: (see Appendix 2)
- B. Match with the specification:

To meet the requirements of the second part of the project, the model was extended to handle a non-linear problem, which makes classical MILP optimization insufficient on its own. Introducing an algorithm that couples a metaheuristic (PSO) with a linear solver (MILP) lets us validate the specification point by point, in particular the minimization of the cost per kWh distributed. The algorithm gets past the non-linearity of the objective function through the PSO and the MILP adjustments, computing the minimum-cost network for each proposed configuration. We should have found the best solution for only 15 nodes, provided that the PSO converged to a global minimum.

The visualization and equity parts are unchanged from part one. However, we now have data on the gas share, the recovered energy and the subsidies, which allows us to include and assess the environmental dimension.

3. System design: technical choices

In both parts of the project, the system follows a nominal scenario structured in three phases to guarantee accurate results:

- **Initialization and data collection:** the user imports the map and thermal data of the buildings in the target area, as well as the source data.
- **Computation and balancing:** the system processes the data and generates a network architecture that aims for the minimum cost per kWh while making sure the constraints are respected.
- **Visualization:** finally, the system generates a clear layout plan. The user reviews the result and validates the construction (final approval).

In part one, we translate the network constraints into mathematical equations, together with an objective function to minimize. For the implementation, we use the Python library PuLP, since the resulting problem is linear.

In part two, we proceed in the same way, except that the resulting mathematical model is no longer linear, hence the need for a metaheuristic. An objective function is computed by a MILP function that takes as input the source number, the installed power and the estimated gas share. Then a PSO, whose parameters are values of installed power and estimated gas share, runs up to 30 iterations of the MILP on a swarm of 15 particles. As described above, the score function takes into account the MILP result and the error on the gas share. To avoid excessive computation time, we only ran our code on 15 nodes. Since the PSO picks its initial values at random, and to prevent the constraint requiring at least 40% of the buildings to be connected from stopping the MILP from finding the optimal solution, we chose (after checking the MILP results) a minimum installed power of 850 kW.

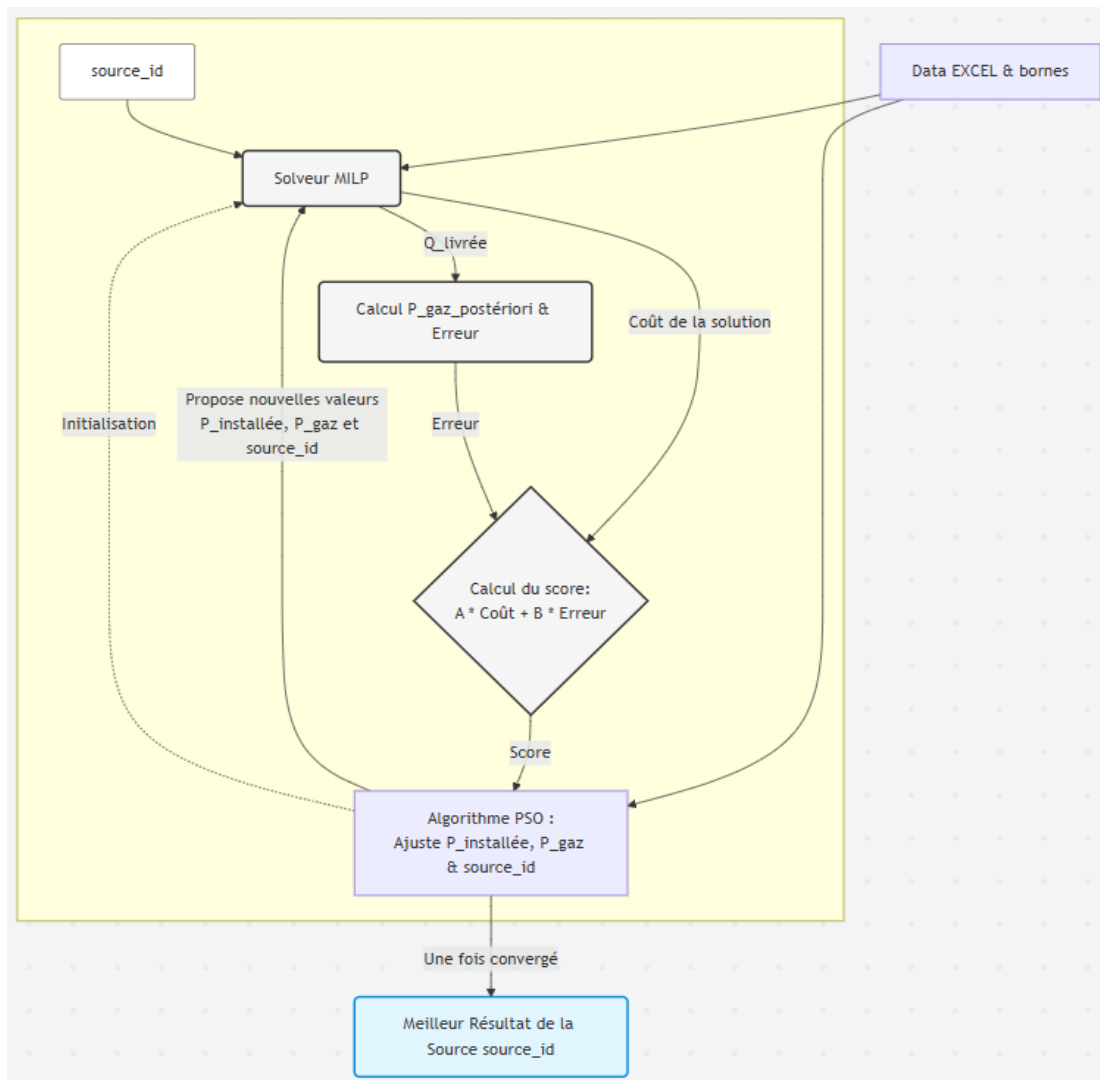


Diagram 1. How the chosen algorithm works

4. Results

4.1 Part one:

```
Status: Optimal
Total Cost = -708531.9315355007
built[0][6] = 1.0
incoming_heat[0][6] = 146.2173
outcoming_heat[0][6] = 0.0
built[1][16] = 1.0
incoming_heat[1][16] = 125.36003
outcoming_heat[1][16] = 0.0
built[3][2] = 1.0
incoming_heat[3][2] = 176.96725
outcoming_heat[3][2] = 0.0
built[3][4] = 1.0
incoming_heat[3][4] = 1280.4137
outcoming_heat[3][4] = 924.41054
built[3][8] = 1.0
incoming_heat[3][8] = 121.63482
outcoming_heat[3][8] = 0.0
built[3][14] = 1.0
incoming_heat[3][14] = 583.60038
outcoming_heat[3][14] = 462.11656
built[3][17] = 1.0
incoming_heat[3][17] = 604.56828
outcoming_heat[3][17] = 371.27812
built[3][19] = 1.0
incoming_heat[3][19] = 707.74263
outcoming_heat[3][19] = 449.08726
built[4][9] = 1.0
incoming_heat[4][9] = 924.41054
outcoming_heat[4][9] = 673.38782
built[9][5] = 1.0
incoming_heat[9][5] = 221.01744
outcoming_heat[9][5] = 0.0
```

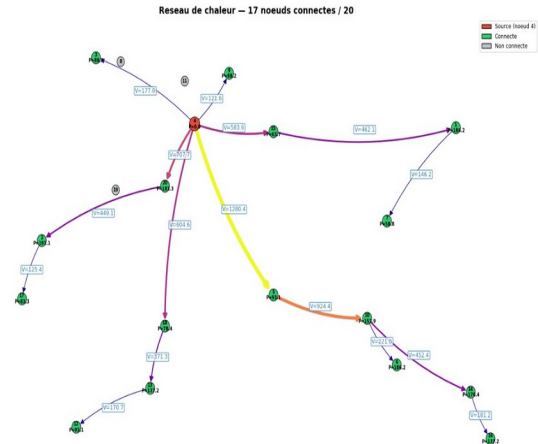


Figure 1. Screenshot showing an extract of the result obtained in the terminal and the corresponding graph for the minimum-distribution equity constraint (see Appendix 2)

```

Status: Optimal
Total Cost = -711219.3097021646
built[0][6] = 1.0
incoming_heat[0][6] = 1127.8298
outcoming_heat[0][6] = 954.33851
built[1][16] = 1.0
incoming_heat[1][16] = 125.36003
outcoming_heat[1][16] = 0.0
built[3][0] = 1.0
incoming_heat[3][0] = 1594.5692
outcoming_heat[3][0] = 1127.8298
built[3][7] = 1.0
incoming_heat[3][7] = 244.57614
outcoming_heat[3][7] = 109.18063
built[3][8] = 1.0
incoming_heat[3][8] = 121.63482
outcoming_heat[3][8] = 0.0
built[3][14] = 1.0
incoming_heat[3][14] = 113.91029
outcoming_heat[3][14] = 0.0
built[3][18] = 1.0
incoming_heat[3][18] = 1422.004
outcoming_heat[3][18] = 1225.3224
built[6][9] = 1.0
incoming_heat[6][9] = 954.33851
outcoming_heat[6][9] = 673.38782
built[7][2] = 1.0
incoming_heat[7][2] = 109.18063
outcoming_heat[7][2] = 0.0
built[9][5] = 1.0
incoming_heat[9][5] = 221.01744
outcoming_heat[9][5] = 0.0
built[9][13] = 1.0
incoming_heat[9][13] = 452.37038

```

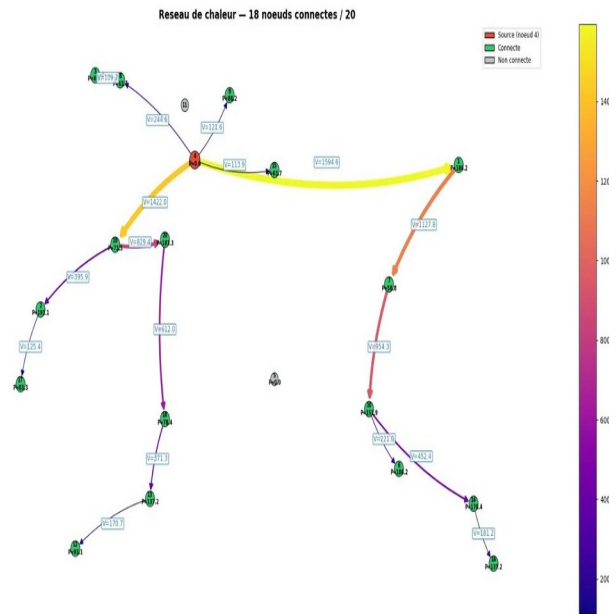


Figure 2. Screenshot showing an extract of the result obtained in the terminal and the corresponding graph for the minimum-demand equity constraint (see Appendix 2)

```

Status: Optimal
Total Cost = -711219.3097021646
built[0][6] = 1.0
incoming_heat[0][6] = 1127.8298
outcoming_heat[0][6] = 954.33851
built[1][16] = 1.0
incoming_heat[1][16] = 125.36003
outcoming_heat[1][16] = 0.0
built[3][0] = 1.0
incoming_heat[3][0] = 1594.5692
outcoming_heat[3][0] = 1127.8298
built[3][7] = 1.0
incoming_heat[3][7] = 244.57614
outcoming_heat[3][7] = 109.18063
built[3][8] = 1.0
incoming_heat[3][8] = 121.63482
outcoming_heat[3][8] = 0.0
built[3][14] = 1.0
incoming_heat[3][14] = 113.91029
outcoming_heat[3][14] = 0.0
built[3][18] = 1.0
incoming_heat[3][18] = 1422.004
outcoming_heat[3][18] = 1225.3224
built[6][9] = 1.0
incoming_heat[6][9] = 954.33851
outcoming_heat[6][9] = 673.38782
built[7][2] = 1.0
incoming_heat[7][2] = 109.18063
outcoming_heat[7][2] = 0.0
built[9][5] = 1.0
incoming_heat[9][5] = 221.01744
outcoming_heat[9][5] = 0.0
built[9][13] = 1.0
incoming_heat[9][13] = 452.37038

```

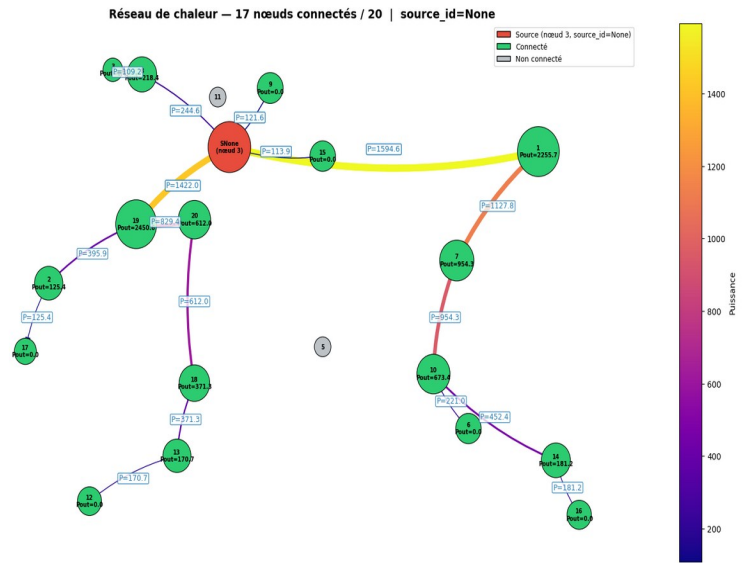


Figure 3. Screenshot showing an extract of the result obtained in the terminal and the corresponding graph for the minimum-surface equity constraint. NB: 18 nodes connected, not 17 (see Appendix 2)

4.2 Part two:

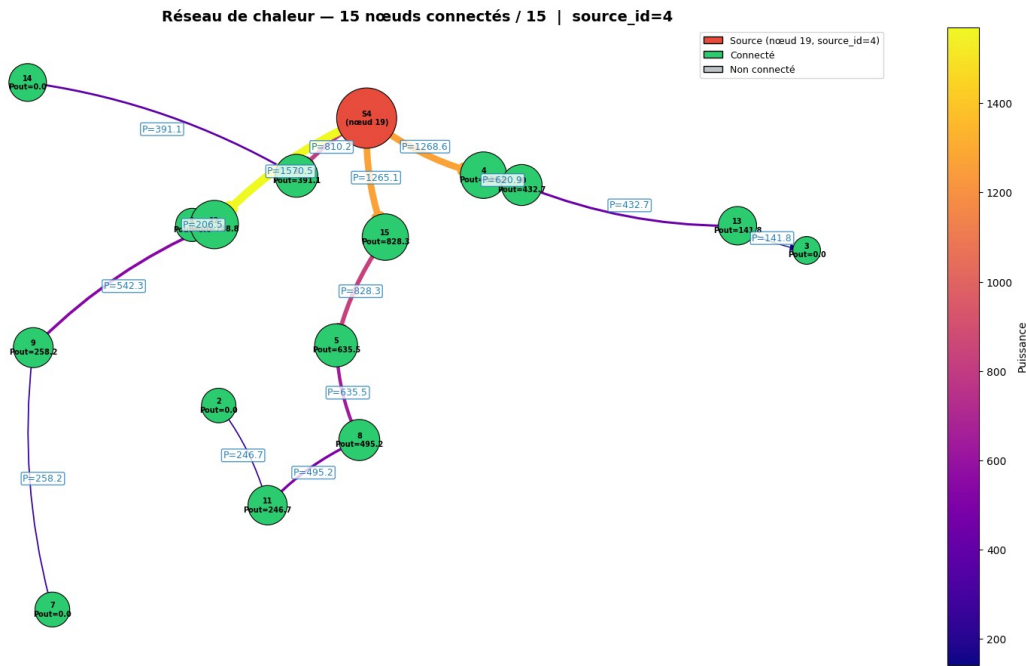


Figure 4. Screenshot showing the graph obtained with a metaheuristic algorithm (PSO)

Results printed in the terminal:

- **Optimal source:** S5 (`source_id` = 4)
- **Optimal installed power:** 4,931.65 kW
- **Optimal gas share:** 0.1560 (i.e. 15.60%)
- **Minimum score found:** -€725,096.2187
- **Final MILP status:** Optimal
- **Final MILP cost:** -€725,524.6470

In the end, our algorithm produced a balanced heating network after about 20 hours of computation. The program only ran on the first 15 nodes to reduce the computation time. The results show that the overall cost of this solution remains close to that of the previous situation.

The result is completely wrong: the installed power is huge compared with the final gas share. This comes from a calculation error in a function that computes the gas share. The gas share is actually 1, which necessarily makes the result wrong: the subsidy no longer applies and the operating cost increases. Because the computation time is extremely long, we did not have the opportunity to rerun the program with the intended fix.

5. Table of scientific/technical deliverables, with their description, format and due dates:

Deliverable	Status	Description	Sub-tasks	Category	Deadline	Duration	Planned margins	People in charge
Team contract	Not graded	Document setting out all the rules that frame the teamwork	-	organizational	03/02/2026	1	-	Whole team
Scoping document	Graded	Document that reports the initial state of your project along several axes: scientific/technical and organizational (team, project).	-Introduction -Specification -Work plan -Team organization -Risk analysis -Conclusion	organizational/ technical/scientific	01/03/2026	15	5	Whole team
Individual reflection questionnaire	Not graded	Initial reflection for the "Improve your professional stance" workshop.	-	organizational	29/03/2026	1	-	Individual task
Impact analysis	Not graded	Reflection on the impacts of the designed system	-	social/ environmental	29/03/2026	1	-	Modelling sub-team
Project and team review	Graded	Oral presentation to the supervisor. Progress review on: scientific/technical contribution, project management and team functioning	-Objectives, updates to the deliverables table and justification of deviations -Project progress: design, development, tests/validation, justification of the choices made, short-	organizational/ technical/scientific	07/04/2026	10	4	Split into a technical part (computing sub-team) and an organizational and presentation part (modelling sub-team)

			term outlook, difficulties, and fallback/backup/workaround solutions					
Analysis of the team dynamics	Not graded	Reading the email sent, team discussion and summary	-	organizational	07/04/2026 before 23:55	1	1	Modelling sub-team
Life cycle analysis (LCA)	Graded	The goal is to introduce the notions of life cycle analysis and life cycle thinking.	-. -Life cycle analysis of the proposed system, the subject of our PRONTO project.	social/ environmental	26/04/2026	3	1	Modelling sub-team
Societal transitions	Graded	The goal is to put students in a situation of controversy that is complex, uncertain, ambiguous and volatile (VUCA).	-Writing a story of a just transition, achieved in 2035 and 2050, describing: - the factors that made this transition easier - how things work after the transition, at the chosen date	social/ environmental	22/05/2026	3	1	Modelling sub-team

Project report	Graded	Document that reports the final state of your project along several axes: scientific/technical and organizational (team, project).	-In early May, the team holds a dedicated meeting with the technical supervisor about the report and presents at least a detailed outline of it. -Introduction: context, problem statement, objectives and structure of the document. -Specification: reminder of the expected functions. -Design & Development: justification of the technical choices, architecture, implementation details and unit tests. -Integration & Validation: system assembly, full tests and demonstration scenarios. -Management & Closure: deliverables table, areas for improvement and conclusion with outlook. -Lessons learned: analysis of the planning (Gantt, deviations), risk management and review of the teamwork.	organizational/ technical/scientific	26/05/2026	10-12	4	Split into a technical part (computing sub-team) and an organizational and presentation part (modelling sub-team)
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Flash oral presentation of the project	CANCELLED	Present the project and its results in a simple and concise way.	-Introduction: context and problem statement -Approach and results: what works, demonstration of the technical success. -Outlook: limitations, possible next steps and improvements. -Engineer profile (CAR method): 1 summary CV slide (Context, Action, Result) as relevant bullet points.	organizational/technical/scientific	02/06/2026	8	3	Computing sub-team
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Table 1. Table of scientific and technical deliverables

6. Areas for improvement

To improve the performance of the hybrid optimization, several changes could be made to the solver, the code architecture and the PSO settings. An analysis of the nodes with more than 40% heat losses would also make it possible to check whether the location of source 4 is relevant.

On the software side, it is better to instantiate the `multiprocessing.Pool` object once, before the PSO loop starts, to avoid the cost of creating the processes again and again. In addition, the ten candidate sources could be processed in parallel rather than sequentially. Here, it is the MILPs that were parallelized.

7. Lessons learned on project management

7.1. Time planning:

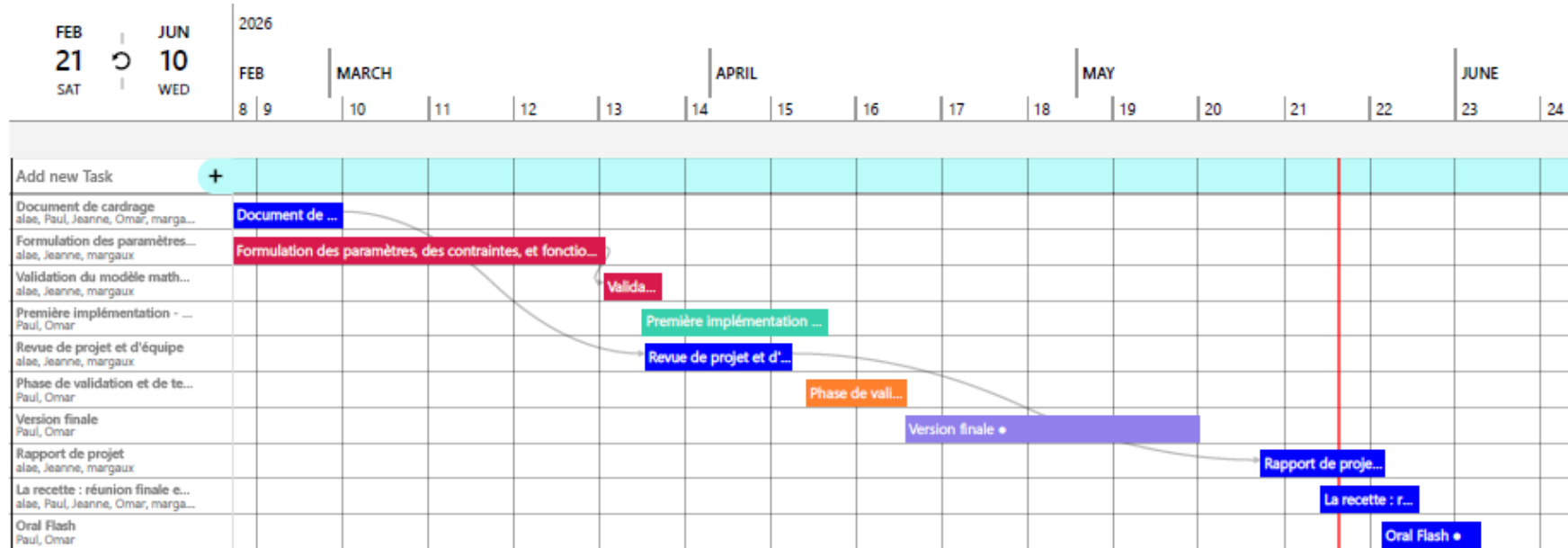


Figure 4. Initial and mid-project Gantt chart

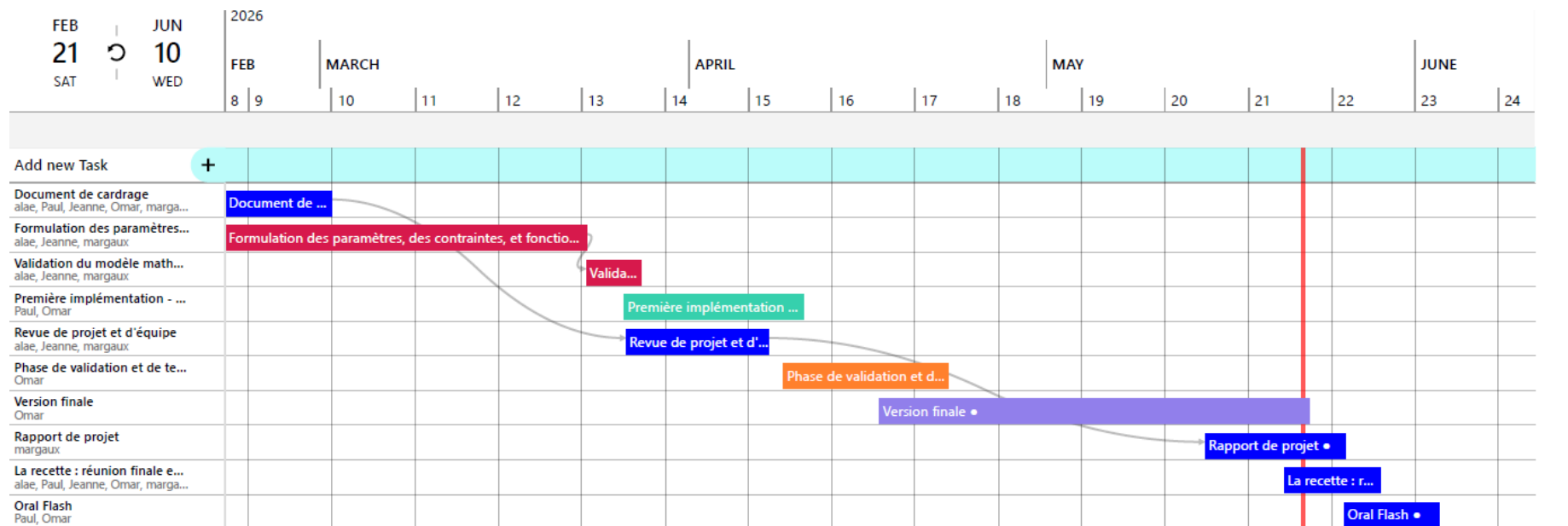


Figure 5. Final Gantt chart

We note that validating part one took longer than planned because of the debugging problems we ran into, detailed in the unit tests section. This delayed the start of part two, and therefore the implementation of the final version of the system.

However, we started the deliverables earlier to make up for this delay, and we reorganized the team so that the computing sub-team could focus fully on designing the system according to the constraints of part two.

7.2. Risk management

Problem	Probability	Risk	Solution
Absence of a member (occasional)	High	Limited	Give notice of absences that affect the group so that a solution can be found Appoint substitutes
Absence of a member (frequent or long)	Limited	High	Appoint substitute leads so that the tasks get done Everyone able to do every task Plan buffer periods
Impossible to find a common time slot	High	Moderate	Use digital tools (video calls) Book fixed time slots Hold the meeting with the people available
IT failure/problem (or other tools)	Moderate	High	Back up all content online and on each of our devices
Lack of involvement of a member	High	High	Appoint substitutes Talk about one's motivation Let people choose their tasks
Conflict	High	High	Communicate Respect each other Set rules Appoint a lead (outside the conflict if possible) Discuss it with a supervisor.
Poor estimate of the schedule	High	High	Plan large margins Plan buffer periods
Poor split of the tasks	Limited	Limited	Gantt chart Regular meetings
Underestimating the difficulty of the project	Limited	Limited	Extra time
Misunderstanding of the assessment criteria/the need	High	High	Discussion with the supervisor

Table 2. Summary of the risks

8. Lessons learned on teamwork

Overall, our work went very well thanks to an effective split of the tasks. We divided the group in two: Alae, Jeanne and Margaux handled the mathematical modelling, while Paul and Omar handled the computer implementation. This split was based on our skills and interests.

In addition, Alae was responsible for running the project smoothly. She made sure all tasks were properly assigned and done on time, and she was in charge of the Gantt chart. Margaux was also in charge of answering our supervisor's emails and writing to him when needed. One of the most instructive aspects of our teamwork on this project was the flexibility we brought to our organization. At the start of the semester, we had planned a weekly meeting every Monday morning, to move the project forward outside the time slots allocated to it in our timetable and to review its progress regularly. Although these meetings proved very useful on many occasions, we quickly realized that they were not always necessary. We therefore gradually adopted a more flexible organization than initially planned, meeting on Mondays only when it was useful. This is probably one of the main lessons we will take away: contrary to what we thought at first, a flexible organization made us more efficient while avoiding unnecessary losses of time and energy.

9. Conclusion

The project “**Chaud pour décider ?**” (**Hot to Decide?**) allowed us to address a concrete problem linked to today's challenges of energy transition and sustainable land planning. Through this work, we developed a decision-support tool able to propose an optimized architecture for an urban heating network, taking technical, economic and social constraints into account.

Part one of the project led us to build a linear mathematical model to optimize a network fed by a single heat source. This phase allowed us to translate the constraints of the specification into mathematical equations and then implement them to obtain consistent and usable results. Part two led us to work on a more complex problem with several heat sources, which required a suitable metaheuristic approach.

Beyond the technical results, this project was a particularly rewarding human and organizational experience. We learned to work as a team, to split tasks efficiently according to everyone's skills, and to adapt our organization when facing difficulties. The debugging phases and the rework of the model in particular showed us the importance of scientific rigour, communication and collective questioning when running a technical project.

Finally, several improvements could be considered to enrich the system. In particular, it would be possible to include more environmental and regulatory constraints, to improve the accuracy of the data used, or to go further with the optimization methods used in part two. A more developed user interface could also make the tool easier to use for non-specialist decision-makers.

This project thus allowed us to use and develop scientific, technical and organizational skills, while making us more aware of today's energy and environmental issues.

Bibliography:

- [1] “Guide Étudiants” (Student guide), course document available on Moodle.
- [2] Large-Scale District Heating Network Optimization, by Johannes Dorfner and Thomas Hamacher, IEEE Transactions on Smart Grid, vol. 5, no. 4, July 2014. Document available on Moodle.
- [3] Particle swarm optimization, Wikipedia, the free encyclopedia. Available at:

[Wikipedia – Particle swarm optimization.](#)

[4] “Consignes pour la deuxième phase du projet” (Instructions for the second phase of the project), course document available on Moodle.

Appendix 1

MARGAUX

Tell me about a situation in this project where you faced a problem that had several possible solutions.

1. Context

The first major step of our project was to build a mathematical model of the problem, in order to define precisely the constraints and the objective function needed to install the network. This work was done within the mathematical modelling team, made up of Jeanne, Alae and myself.

Once this theoretical phase was over, we started a second essential step: implementing these constraints and the objective function in the code. That is when we ran into a major difficulty. Despite several attempts, the program did not produce the expected results, and some constraints seemed incompatible at run time.

We had two options: keep looking for the source of the problem by debugging the code only, or completely rework the mathematical model to check that the initial equations and assumptions were correct. It was an important decision, because reworking the theoretical part meant a lot of work, but it could reveal deeper errors in our reasoning.

2. Action

After many debugging attempts that did not solve the problem satisfactorily, we finally decided to redo the whole mathematical part from the beginning. We chose this approach to make sure the theoretical foundations of the project were consistent before continuing the software development.

To do so, each of us individually reread the scientific documentation and the technical resources provided by our supervisor. We then compared our analyses and pooled our remarks within the mathematical modelling team. This collaborative work allowed us to review every constraint and to check the assumptions used and the consistency of the equations in the model.

Once the corrections were identified, we updated our model and presented our work to our supervisor again to get it validated before continuing the implementation.

3. Result

This complete rework of the model allowed us to identify several errors and inconsistencies in the initial mathematical formulation. After we corrected them, our supervisor validated the new version of the model, which confirmed that our reasoning was now consistent.

Thanks to these changes, the implementation then worked correctly and the constraints were taken into account without any major error in the program. This experience also taught us how important it is to check the theoretical part rigorously before trying to solve a problem through code alone, and the value of teamwork in solving complex technical difficulties.

JEANNE

Tell me about a situation in this project where you faced a problem that had several possible solutions.

1. Context

During the first phase of our project, I joined the sub-group in charge of the mathematical modelling, made up of Alae, Margaux and myself. Our objective was to build a mathematical model that correctly represents how our system works and its constraints. This step was essential, because the rest of the project depended on how reliable this model was.

For this work, we had a technical guide written in English that explained the steps needed to establish the expected mathematical functions. We relied on this documentation to build our equations and our model. However, after a check by our supervising teacher, Timothé Gronier, we found that some of the formulas we had written contained errors.

2. Action

To correct these inconsistencies, the three of us, Alae, Margaux and I, decided to rework the whole model. We went through each formula one by one with the help of the guide provided, while carefully checking the calculations and the assumptions used.

We also compared our reasoning to validate each step collectively and avoid new errors. This way of working allowed us to progress in a more rigorous and organized way.

3. Result

This thorough review allowed us to identify and gradually correct the errors in our initial model. At the end of this checking phase, we had a consistent mathematical model, validated by our supervisor.

This experience showed us the importance of scientific rigour, of checking calculations and of teamwork in the success of a technical project.

ALAE

Tell me about a time in this project when you took a risk, made a mistake or failed.

1. Context:

At the start of the project, we had to define a mathematical model to make the implementation easier. We therefore split into two sub-teams, and I was part of the sub-team in charge of the mathematical modelling.

2. Actions:

We had to read scientific articles in English and French, then discuss with my colleagues Jeanne and Margaux to check our understanding of the documents. This work therefore required good communication within the team. We then presented our progress to our supervisor. The first meeting with him revealed several errors, due either to a misunderstanding of the texts provided or to a lack of attention. In particular, we were stuck on a constraint that could be written in three different ways, and on the expression of the heat cost. The next two meetings allowed us to make corrections, which we took into account to reach a consistent final model.

However, one small error was only spotted during the implementation with Paul. It was a source of frustration for the team, all the more so as it only took correcting the indices of a variable.

3. Results:

Thanks to our regular exchanges with our supervisor and our communication with the rest of the team, we were able to overcome the difficulties of the mathematical modelling. This allowed us to move forward efficiently and get our model working.

PAUL

Describe a situation where you took the lead during the project.

1. Context:

At the end of part one, when we started looking at the results, there were many inconsistencies. After rereading the code against the mathematical model several times, I found no copying error. So either I had made a mistake in the code because of a misunderstanding, or there was an error in the mathematical model. But the model had been built jointly by the three members of the optimization team and reviewed by the supervisor, so it was unlikely to be wrong.

2. Action

I started by checking how I extracted the data from the Excel file we had been given: no test had been done. A few errors were corrected, in particular thanks to Omar. Then, following the supervisor's advice, I tested the constraints one by one. Some did not seem to apply. By drawing the graphs the code produced after the optimization, I was able to spot the constraints that looked wrong. As my understanding of the mathematical model was limited, at the next session I discussed the errors I had found with the modelling team: some constraints did not seem to work, both in the program and on outside examples. With Alae, I went back over the constraints causing errors and we found very simple index errors that nevertheless changed the results deeply. We then made an appointment with the supervisor to present our result. It was still not optimal. After reviewing it, he helped us identify the last errors, which allowed us to obtain the optimal result.

3. Result

We finally obtained a working result. What this incident shows is that, although it is not possible to do the whole project alone in the time available, teamwork with constant questioning of our results is necessary. Every member must understand what the rest of the team has done and why.

OMAR

Tell me about a situation in this project where you faced a problem that had several possible solutions.

1. Context

During one phase of the project, we had to present our progress to our supervising teacher at a meeting at school. The most important part to show was the visualization, because it made our work concrete. The problem was that the only code that worked with this visualization was on my personal computer. That same day, a few minutes before the meeting, my network card stopped working. I no longer had a connection, while we needed to run the project properly and prepare the demonstration.

2. Action

I first tried to fix the problem directly on my computer. I started with the simple solutions: restarting the PC, disabling and re-enabling the connection, checking the network settings. As that was not enough, I tried more advanced solutions, including resetting the network configuration from the terminal. Still, the connection did not come back, and I realized that continuing to insist would make us lose too much time.

So I chose another solution: moving the project to another computer. I borrowed a USB stick, copied my work onto it, then installed it on Paul's and Alae's computers. I then quickly installed the missing libraries and dependencies so that the code could run properly on their machines. We had to improvise fast, because a first presentation had already been given without the visualization, which made our progress less clear.

3. **Result**

About 45 minutes later, we managed to get the code and the visualization working on another computer. We were therefore able to present our progress fully to the teacher. This situation taught me that in a project, you have to stay calm when something unexpected happens, try several solutions, but also change strategy when a solution takes too long. It also showed me the importance of having backups and not depending on a single computer for an important presentation.